M2 · ECE Module 2, semiconductor devices, amplifiers and communication systems

Semiconductors, transistors, op-amps & communications

What PN diodes, Zener regulators, rectifiers, BJTs, and MOSFETs do, how operational amplifiers work internally, and how EM spectrum, cellular, satellite, and radar systems operate.

○ Mark as done ○ Highlight Quiz this module · FAT PYQs

All syllabus sections & lecture slide diagrams included · Ticks itself when every tab is read

PN & Zener Diodes ✓ Rectifiers BJT & MOSFET Op-Amp (IC 741) EM & Comm Systems Cellular, Satellite & Radar

Syllabus Order: 1. PN Diode • 2. Zener • 3. BJT • 4. MOSFET • 5. Rectifiers • 6. Op-Amp • 7. EM & Comm • 8. Cellular • 9. Satellite • 10. Radar
Sources: VIT CAT-2 E1 Paper • VIT CAT-2 Official Answer Key

M2T1 Module 2, topic 1

Semiconductor Physics & PN Junction Diode (Biasing & V-I Characteristics)

Concepts & Working ✓ Key Equations Slide Diagrams Solved PYQ

Overview & context	Why is **Silicon (Si, Atomic No. 14, 4 valence electrons, Bandgap E_g = 1.1 eV)** the heart of electronics? Pure ("intrinsic") silicon is a poor conductor at room temperature, so we inject a tiny pinch of impurity atoms (**Doping**) to create two building blocks (VIT BAEEE101 Syllabus):
N-Type Semiconductor (Pentavalent Doping — Group 15)	Impurity has **5 valence electrons**. Four form covalent bonds with Silicon; the 5th is free to conduct! **Majority carriers = Electrons (−); Minority carriers = Holes (+)**.
P-Type Semiconductor (Trivalent Doping — Group 13)	Impurity has only **3 valence electrons**, creating an empty positive seat called a **Hole (+)**. **Majority carriers = Holes (+); Minority carriers = Electrons (−)**.
How the PN Junction & Depletion Region Form	When P and N join, electrons near the border diffuse into P-side holes, leaving behind uncovered **immobile positive donor ions (+ on N-side)** and **immobile negative acceptor ions (− on P-side)**. This charge-free wall is the **Depletion Region**, creating a built-in barrier voltage (tollbooth fee): **V_0 = 0.7 V for Silicon (Si)** and **0.3 V for Germanium (Ge)**.
Forward Bias (P → +ve, N → −ve, Figure 2.1-1 Top)	External battery pushes majority carriers across the border, **narrowing the depletion layer**. Once V ≥ 0.7 V (Knee/Cut-in voltage V_K), current shoots up exponentially in milliamperes (mA): **$I = I_0(e^{V / \eta V_T} - 1)$** (Acts as a **Closed ON Switch**).
Reverse Bias (P → −ve, N → +ve, Figure 2.1-1 Bottom)	External battery pulls majority carriers away from the junction, **widening the depletion layer**. Majority current is blocked (Acts as an **Open OFF Switch**); only a

tiny microampere (μA) reverse saturation current I_0 flows due to thermally generated minority carriers.

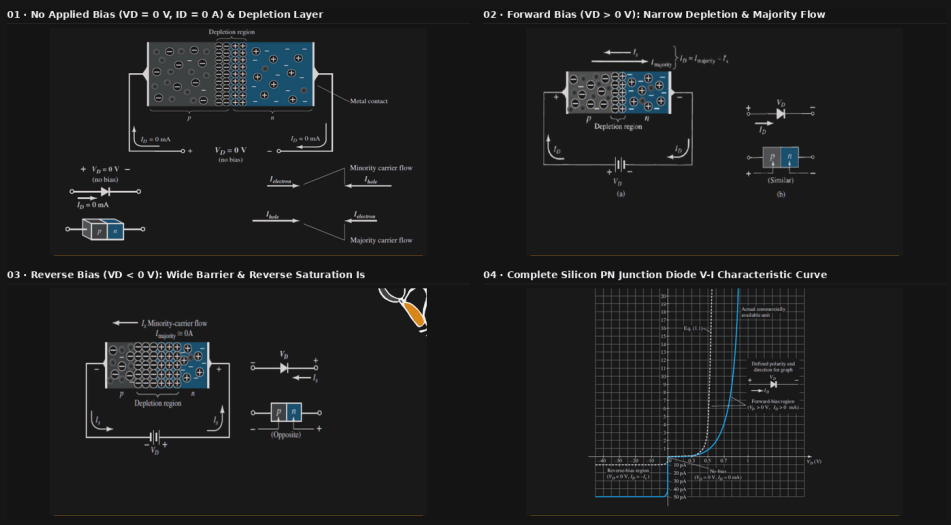


Figure 2.1B (Extracted from VIT Lecture Slides): (a) No Applied Bias (V_D = 0 V, I_D = 0 A), (b) Forward Bias (V_D > 0 V: Majority diffusion $I_D = I_{majority} - I_s$), (c) Reverse Bias (V_D < 0 V: Wide depletion barrier & minority saturation current I_s), (d) Complete Silicon PN Diode V-I Curve.

Complete Slide Theory

Instant of Joining & No Applied Bias (VD = 0 V)

At the instant N-type and P-type materials are joined, free electrons and holes immediately adjacent to the junction cross over and recombine, leaving behind uncovered immobile positive donor ions (+ on N-side) and negative acceptor ions (– on P-side). This carrier-free zone is the **Depletion Region (Space-Charge Layer)**. With no external voltage applied (V_D = 0 V), any minority carrier thermally generated near the junction that drifts across is exactly balanced by an opposite majority carrier overcoming the barrier, so net current is strictly **I_D = 0 mA** (acting like an open circuit).

Forward Bias (VD > 0 V — "ON" State)

Established by connecting the positive battery terminal to P-type and negative terminal to N-type. The external electric field opposes the internal barrier potential, pressuring majority electrons in N and holes in P toward the junction. This reduces the barrier height and **narrows the depletion width** (marked by heavy majority-carrier vectors in the slide diagram). Net diode current is **$I_D = I_{majority} - I_s$**. Above the Knee/Cut-in Voltage (**V_K = 0.7 V for Silicon, 0.3 V for Germanium, 1.2 V for GaAs**), I_D rises exponentially according to **Shockley's Equation: $I_D = I_s \cdot (e^{V_D / (n \cdot V_T)} - 1)$**, where thermal voltage $V_T = kT/q \approx 26$ mV at 300 K and ideality factor $n \in [1, 2]$. Notice on the slide V-I axis that the forward voltage scale is in *tenths of a volt (< 1 V)* while forward current is in *milliamperes (mA)*!

Reverse Bias (VD < 0 V — "OFF" State)

Established by connecting the positive terminal to N-type and negative terminal to P-type. Free electrons are drawn away from the junction toward the + terminal and holes toward the – terminal, **widening the depletion region** until the barrier potential equals the applied reverse voltage V_D. Majority current drops to zero ($I_{majority}$ = 0); however, thermally generated minority carriers (electrons in P, holes in N) are swept across the junction, producing a tiny, voltage-independent

	Reverse Saturation Current I_s (measured in μA for Ge and nA for Si; doubles for every 10°C rise in temperature!).
Breakdown Region (VBV) & Zener vs. Avalanche Mechanisms	When reverse voltage reaches Breakdown Voltage V_{BV}, reverse current increases sharply in the vertical direction. **Zener Breakdown (V_Z < 6 V):** Occurs in **heavily doped** PN junctions with a very narrow depletion layer. Even a small reverse voltage (2.4 V to 6 V) creates an intense electric field of **~10^7 V/m (300,000 V/cm)** that causes **direct rupture of covalent bonds** (quantum mechanical tunneling). Has a **Negative Temperature Coefficient** (V_Z decreases as T rises). **Avalanche Breakdown (V_Z > 6 V):** Occurs in **lightly doped** junctions with a wide depletion layer. Thermally generated minority carriers accelerate to high kinetic energy and collide with crystal lattice atoms, knocking loose valence electrons by **Impact Ionization (Carrier Multiplication)** like an avalanche. Has a **Positive Temperature Coefficient** (V_Z increases as T rises).
Zener Shunt Voltage Regulator — Line Regulation & Load Regulation	A Zener diode is operated in reverse breakdown between **$I_{Z(min)}$** (knee current to stay in breakdown) and **$I_{Z(max)} = P_{Z(max)} / V_Z$**. Series resistor R_S drops the excess voltage $V_{in} - V_Z$. **Line Regulation:** Ability to maintain constant $V_O = V_Z$ when R_S and R_L are fixed and input voltage V_{in} fluctuates between $V_{in(min)}$ and $V_{in(max)}$: **Line Regulation = $(\Delta V_O / \Delta V_{in}) \times 100\%$**. **Load Regulation:** Ability to maintain constant $V_O = V_Z$ when V_{in} is fixed and load resistance R_L varies from open-circuit (no load, $I_L = 0 \Rightarrow I_Z$ is maximum!) to full load ($R_{L(min)}$, I_L is maximum $\Rightarrow I_Z$ is minimum!): **Load Regulation = $((V_{NL} - V_{FL}) / V_{FL}) \times 100\%$**.

M2T2 Module 2, topic 2

Zener Diode & Zener Shunt Voltage Regulator

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	An ordinary PN diode burns out if pushed into reverse breakdown. A **Zener Diode** (symbol has a "Z"-shaped cathode bar) is a **heavily doped PN junction** specifically engineered to operate safely and continuously in the **Reverse Breakdown Region** without damage (VIT CAT-2 Official Answer Key)!
Zener Breakdown (< 5 V) vs. Avalanche Breakdown (> 6 V)	In *Zener Breakdown* (heavily doped, ultra-thin depletion layer, V_Z < 5 V), an intense electric field (~10^7 V/m) directly rips covalent bonds apart (quantum tunneling). In *Avalanche Breakdown* (lightly doped, V_Z > 6 V), accelerated minority carriers collide with crystal atoms and knock loose an avalanche of electrons (impact ionization).
How the Zener Shunt Voltage Regulator Works (Figure 2.1-3)	Connected in **Reverse Bias in parallel (shunt) across the load R_L**. As long as input voltage $V_{in} > V_Z$, the Zener locks the output voltage across R_L at a rock-steady **V_L**

= V_Z! If V_{in} rises or load current I_L drops, the excess current automatically diverts through the Zener (between I_{Zmin} and I_{Zmax}) so V_L never changes!

4 Equations for Every Zener Regulator Numerical

(1) Series resistor drop: $I_S = (V_{in} - V_Z) / R_S$. (2) Load current: $I_L = V_Z / R_L$. (3) KCL at top node: $I_Z = I_S - I_L$. (4) Zener power dissipation: $P_Z = V_Z \cdot I_Z$.

M2T3 Module 2, topic 3

Bipolar Junction Transistor (BJT — Structure, Modes, CB/CE/CC & Curves)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context

A **BJT (NPN or PNP)** is a 3-terminal, 2-junction **current-controlled device** ("bipolar" because both electrons and holes conduct). Think of it as a **water tap** where a tiny microampere trickle into the **Base (I_B)** opens a floodgate of milliamperes from **Collector to Emitter ($I_C = \beta \cdot I_B$)**!

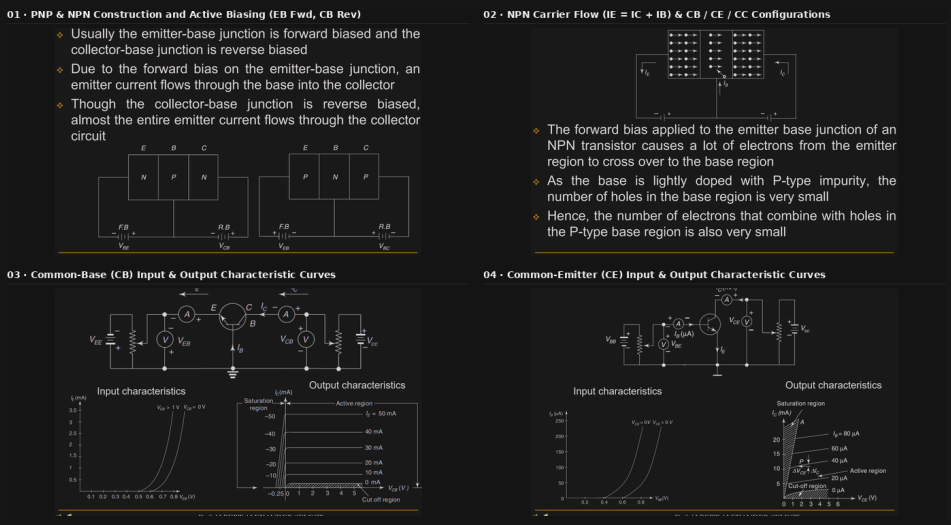


Figure 2.3B (Extracted from VIT Lecture Slides): (a) PNP & NPN Construction and Active Biasing (EB Forward, CB Reverse), (b) NPN Minority/Majority Carrier Flow ($I_E = I_C + I_B$), (c) Common-Base (CB) Input & Output Curves, (d) Common-Emitter (CE) Input & Output Curves.

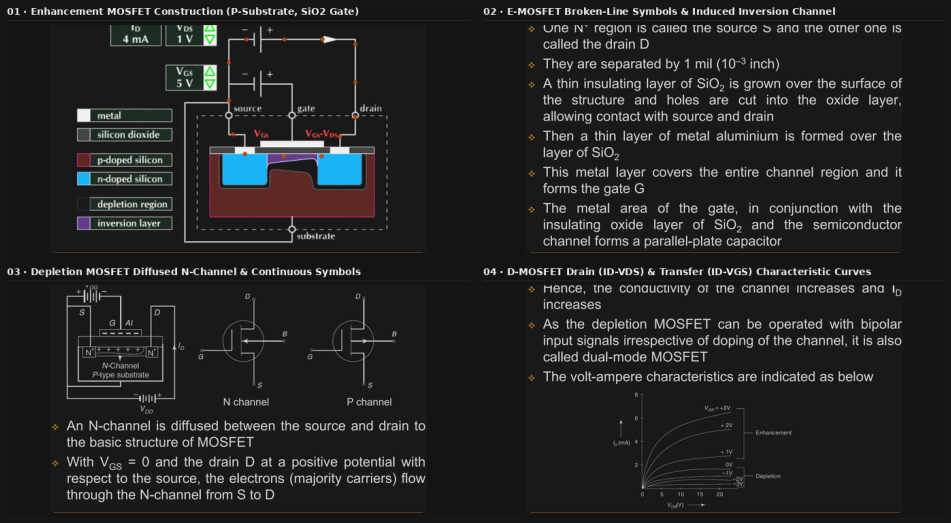


Figure 2.3C (Extracted from VIT Lecture Slides): (a) Enhancement MOSFET Construction (P-Substrate, 1-mil Channel, SiO_2 Gate), (b) E-MOSFET Broken-Line Symbols, (c) Depletion MOSFET Diffused N-Channel & Symbols, (d) D-MOSFET Drain (I_D–V_{DS}) & Transfer (I_D–V_{GS}) Curves.

Complete Slide Theory	
Historical Evolution (Slide 4 of BJT Deck)	**1904:** J. A. Fleming invented the vacuum-tube diode (rectifier). **1906:** Lee De Forest added a third element (control grid) to create the vacuum-tube triode amplifier. Because vacuum tubes were bulky, fragile, slow to warm up, and consumed high filament power, **William Shockley, Walter Brattain, and John Bardeen** invented the first solid-state point-contact transistor at Bell Telephone Laboratories on **December 23, 1947**.
NPN & PNP Carrier Operation	In an active-biased NPN transistor (Emitter-Base forward biased by V_{EE}, Collector-Base reverse biased by V_{CC}), forward bias causes a massive flood of electrons from the heavily doped N-Emitter to cross into the P-Base. Because the Base is extremely thin and lightly doped, **fewer than 5% of electrons recombine with holes** to form the tiny Base current I_B (in μA), while **>95% of electrons diffuse across into the Collector** and are swept out by $+V_{CC}$ as Collector current I_C (in mA). Fundamental nodal law: $\mathbf{I_E = I_C + I_B}$. (In a PNP transistor, majority carriers are **holes** injected from P-Emitter into N-Base and collected at P-Collector).
Input & Output	**Input & Output Characteristic Curves of CB, CE, and CC Configurations:** **CB Input Curve (I_E vs. V_{EB} at constant V_{CB}):** Resembles a forward-biased PN diode curve; as V_{CB} increases, the curve shifts slightly left because the collector depletion layer widens into the base (Early effect). **CB Output Curve (I_C vs. V_{CB} at constant I_E):** Flat horizontal lines in Active region where $\mathbf{I_C \approx I_E}$ ($\alpha \approx 0.98$); saturation occurs only when V_{CB} becomes slightly forward-biased (< 0 V). **CE Input Curve (I_B vs. V_{BE} at constant V_{CE}):** Forward diode curve with I_B in μA and knee at 0.7 V; increasing V_{CE} narrows the effective base width and slightly reduces recombination I_B. **CE Output Curve (I_C vs. V_{CE} at constant I_B):** Divides into 3 distinct regions: (1) **Cut-off Region** (below $I_B = 0$ line; both junctions reverse biased; $I_C = I_{CEO} \approx 0$, open switch), (2) **Active Region** (central gently sloping region; EB forward, CB reverse; $\mathbf{I_C = \beta \cdot I_B}$ linear amplifier), and (3) **Saturation Region** (steep rise near $V_{CE(sat)} \approx 0.2$ V; both junctions forward biased; closed switch).
Complete Slide Theory	
Why FET is Superior to BJT in IC Design	FET is a **unipolar** (majority-carrier only) **voltage-controlled** device with extremely high input resistance ($\mathbf{10^{10}}$ **to** $\mathbf{10^{15}\ \Omega}$ due to the insulated SiO_2 gate layer), low noise, high thermal stability (no thermal runaway!), and tiny silicon chip area.
Enhancement MOSFET (E-MOSFET) Construction & Operation	Built on a lightly doped P-type substrate into which two heavily doped N^+ regions (Source S and Drain D) are diffused, separated by a channel length of about **1 mil (10^{-3} inch = 25.4 μm)**. A thin dielectric layer of **Silicon Dioxide (SiO_2)** is grown over the surface, and an Aluminium metal Gate (G) plate is deposited on top—forming a **parallel-plate capacitor** where the gate and P-substrate are the plates and SiO_2 is the dielectric!

	At V_{GS} = 0 V, no physical channel exists between S and D (two back-to-back PN junctions block current ⇒ **Normally OFF**; represented by a **broken/dashed vertical channel line** in its schematic symbol). Applying a positive gate voltage **$V_{GS} > V_T$** repels holes deep into the P-substrate and attracts free minority electrons to the SiO_2 interface, creating an induced N-type **Inversion Layer** that connects Source to Drain so drain current I_D flows!
Depletion MOSFET (D-MOSFET) Construction & Dual-Mode Operation	During fabrication, a physical N-channel is diffused right between the N^+ Source and Drain underneath the SiO_2 gate (represented by a **solid continuous vertical channel line**). **At V_{GS} = 0 V:** Applying V_{DS} > 0 immediately drives a large drain current **$I_D = I_{DSS}$** (Drain-to-Source Saturation Current; **Normally ON**). As V_{DS} increases, the voltage drop along the channel makes the drain end more reverse-biased than the source end, creating a **wedge-shaped depletion zone** that pinches off and saturates I_D. **Depletion Mode (V_{GS} < 0 V):** Negative gate voltage repels electrons out of the N-channel and attracts holes from the P-substrate for recombination, depleting the channel of free carriers until I_D = 0 at **Pinch-Off / Cut-Off Voltage $V_{GS(off)}$**. **Enhancement Mode (V_{GS} > 0 V):** Positive gate voltage draws additional electrons into the N-channel, increasing conductivity so **$I_D > I_{DSS}$**! Thus its transfer curve (I_D vs. V_{GS}) spans continuously from negative $V_{GS(off)}$ across zero (I_{DSS}) into positive V_{GS}.
Physical Doping & Size of the 3 Regions	(1) **Emitter (E):** *Heavily doped* and medium size (job: inject millions of charge carriers). (2) **Base (B):** *Extremely thin and lightly doped* (so >98% of carriers zip straight through to the Collector without recombining!). (3) **Collector (C):** *Moderately doped and physically the LARGEST area* (must dissipate all the heat!).
Fundamental BJT Current & Gain Equations	By KCL, **$I_E = I_B + I_C$**. Common-Base current gain: **$\alpha = I_C / I_E$** (always slightly < 1, typically 0.95–0.99). Common-Emitter current gain: **$\beta = I_C / I_B$** (typically 50–300). Exact conversion: **$\beta = \alpha / (1 - \alpha)$** and **$\alpha = \beta / (\beta + 1)$**.
The 3 Operating	**The 3 Operating Regions on the CE Output Curve (Figure 2.2-2):**
Active Region (EB)	**Active Region (EB Junction = Forward Biased; CB Junction = Reverse Biased):** $I_C = \beta I_B$ is flat and controlled purely by I_B → used as a **Linear Amplifier**.
Saturation Region (BOTH EB & CB Junctions = Forward Biased)	$V_{CE(sat)}$ ≈ 0.2 V, maximum current flows → acts as a **Closed ON Digital Switch**.
Cut-Off Region (BOTH EB & CB Junctions = Reverse Biased, IB = 0)	I_C ≈ 0 → acts as an **Open OFF Digital Switch**.
Why Common-Emitter (CE, Figure 2.2-1) Beats CB and CC	CE is the ONLY configuration that provides **BOTH high Voltage Gain AND high Current Gain** (giving huge Power Gain), plus a **180° phase shift** between input and output! (CB has current gain α < 1; CC / Emitter Follower has voltage gain ≈ 1 and is used for impedance matching).

M2T4 Module 2, topic 4

MOSFET (Enhancement & Depletion Structure, Working & BJT Comparison)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context

A **MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor)** is a **unipolar, voltage-controlled device** (only majority carriers conduct). Look at its cross-section in **Figure 2.2-3**:

Why Input Impedance Zin ≈ ∞ (Zero Gate Current IG ≈ 0 A)

A microscopic layer of **Silicon Dioxide (SiO_2 glass insulator)** physically separates the Metal Gate (G) from the P-type substrate! Because Gate and Channel form a parallel-plate capacitor, zero DC current ever enters the Gate.

How N-Channel Enhancement MOSFET (E-MOSFET — Normally OFF) Works

At V_{GS} = 0 V, two back-to-back PN junctions sit between Source and Drain, so **no channel exists (I_D = 0, OFF)**. When we apply a positive Gate-to-Source voltage exceeding the **Threshold Voltage ($V_{GS} > V_{th}$)**, the positive electric field across the SiO_2 glass repels holes downward and attracts free electrons to the surface—forming an **N-type Inversion Channel** that connects Source to Drain!

Depletion MOSFET (D-MOSFET — Normally ON)

Built with a physical N-channel already doped between Source and Drain at the factory. It conducts at V_{GS} = 0 V; applying negative V_{GS} depletes the channel ("Depletion mode"), and positive V_{GS} widens it ("Enhancement mode").

M2T5 Module 2, topic 5

Rectifiers (HWR, Center-Tapped FWR, Bridge FWR) & Capacitor Filter

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context

A **Rectifier** converts bidirectional AC voltage into unidirectional pulsating DC. In FAT exams, you must be able to draw the 3 rectifier circuits + waveforms (**Figure 2.3**) and quote their 7 comparison parameters (VIT CAT-2 E1 Paper).

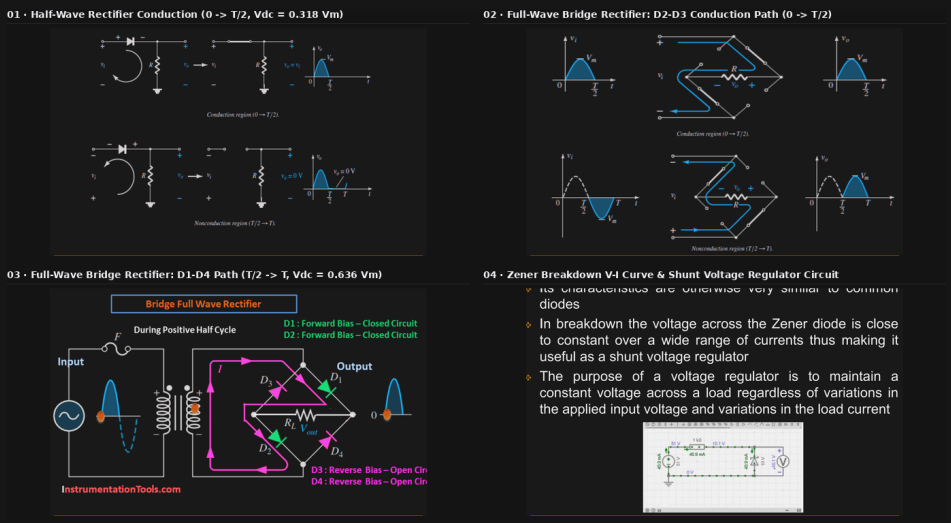


Figure 2.2B (Extracted from VIT Lecture Slides): (a) Half-Wave Rectifier (0→T/2 short-circuit equivalent, V_{dc} = 0.318 V_m), (b) Full-Wave Bridge Rectifier D_2–D_3 conduction (0→T/2), (c) Bridge Rectifier D_1–D_4 conduction (T/2→T, V_{dc} = 0.636 V_m), (d) Zener V-I & Shunt Regulator.

Slide Step-by-Step Rectifier	
Half-Wave Rectifier (HWR)	During **t = 0 → T/2** (positive half-cycle), the diode is forward-biased and replaced by an ideal short circuit (so $v_o = v_i$). During **t = T/2 → T** (negative half-cycle), the diode is reverse-biased and replaced by an open circuit (so v_o = 0 V). Average DC output over one full period T: $\mathbf{V_{dc} = V_m / \pi = 0.318\ V_m}$. Accounting for a practical silicon diode knee voltage V_K = 0.7 V: $\mathbf{V_{dc} = 0.318 \cdot (V_m - V_K)}$, and peak load voltage is $V_{o(max)} = V_m$ − 0.7 V.
Full-Wave Bridge Rectifier (4 Diodes D1, D2, D3, D4)	Improves the DC level by **100% (doubles V_{dc})** without requiring a bulky center-tapped transformer! **Positive Half-Cycle (0 → T/2):** Top AC terminal is (+) and bottom is (−). In the slide bridge topology, diodes **D_2 and D_3 are forward-biased (ON)** while D_1 and D_4 are reverse-biased (OFF), sending current through load resistor R from (+) to (−). **Negative Half-Cycle (T/2 → T):** Top AC terminal is (−) and bottom is (+). Diodes **D_1 and D_4 are forward-biased (ON)** while D_2 and D_3 are OFF, routing current through R in the *exact same direction*! **Average DC Voltage:** Since the area under the curve over one period T is twice that of a half-wave system, $\mathbf{V_{dc} = 2V_m / \pi = 2(0.318\ V_m) = 0.636\ V_m}$. Because two diodes conduct in series during each half-cycle, with practical silicon diodes (V_K = 0.7 V each, total drop $2V_K$ = 1.4 V): $\mathbf{V_{dc} = 0.636 \cdot (V_m - 2V_K)}$!

Parameter	Half-Wave Rectifier (HWR)	Center-Tapped FWR	Bridge Full-Wave Rectifier
Number of Diodes	**1 Diode** (Conducts positive half only)	**2 Diodes** (D_1 on +ve, D_2 on −ve)	**4 Diodes** (D_1,D_3 on +ve; D_2,D_4 on −ve)
DC Output Voltage (V_{dc})	$\mathbf{V_{dc} = V_m / \pi = 0.318\ V_m}$	$\mathbf{V_{dc} = 2V_m / \pi = 0.636\ V_m}$	$\mathbf{V_{dc} = 2V_m / \pi = 0.636\ V_m}$ (Ideal)
RMS Output (V_{rms}) & Efficiency	$\mathbf{V_{rms} = V_m / 2}$ | Max **η = 40.6%**	$\mathbf{V_{rms} = V_m / \sqrt{2}}$ | Max **η = 81.2%**	$\mathbf{V_{rms} = V_m / \sqrt{2}}$ | Max **η = 81.2%**
Ripple Factor (γ) & Ripple Freq	**γ = 1.21** | f_{ripple} = **f_{in} (50 Hz)**	**γ = 0.482** | f_{ripple} = **$2f_{in}$ (100 Hz)**	**γ = 0.482** | f_{ripple} = **$2f_{in}$ (100 Hz)**
Peak Inverse Voltage (PIV)	**PIV = V_m**	**PIV = $2V_m$** (Needs costly transformer)	**PIV = V_m** (2 diodes drop 1.4 V in series)

M2T6 Module 2, topic 6

Operational Amplifiers (IC 741 — Ideal vs. Practical & Closed-Loop Circuits)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	An **Op-Amp (IC 741)** is a direct-coupled, high-gain differential amplifier. In open loop, its gain is massive ($A_{OL} \approx 10^5$), so even microvolts saturate the output to $\pm V_{sat}$. By adding an external feedback resistor **R_f** from V_{out} back to the **Inverting**

(−) input (Negative Feedback), we unlock the **Virtual Short / Virtual Ground Rule ($V_- = V_+$ and $I_- = I_+ = 0$)** (VIT CAT-2 Key)!

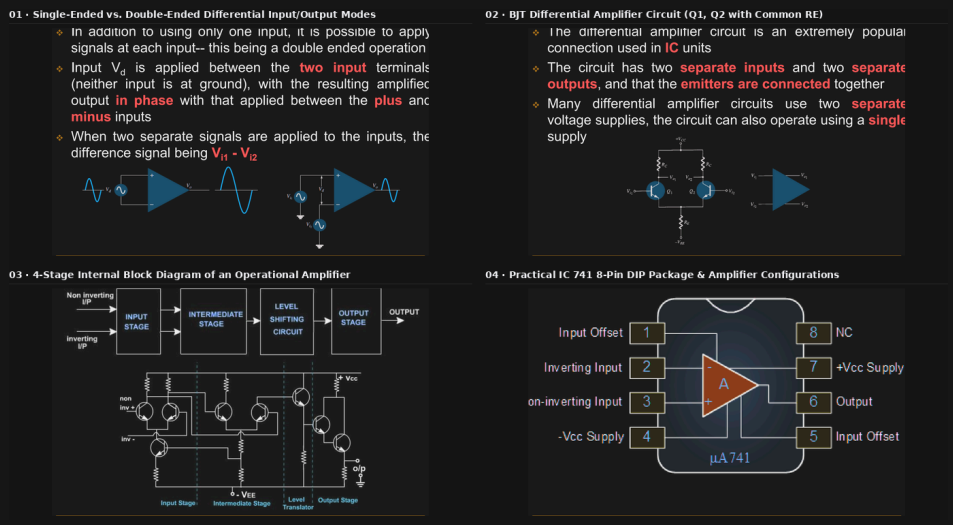


Figure 2.4B (Extracted from VIT Lecture Slides): (a) Single-Ended vs. Double-Ended Differential Input/Output Modes, (b) BJT Differential Amplifier Circuit (Q_1, Q_2, R_E), (c) 4-Stage Internal Block Diagram of an Op-Amp, (d) Practical IC 741 Pinout & Amplifier Circuits.

Complete Slide Theory

Single-Ended Input Mode

Signal V_i is applied to one input terminal while the other input is grounded (0 V). If applied to (+) Non-Inverting input, output V_o is in-phase with V_i; if applied to (−) Inverting input, V_o is 180° out-of-phase.

Double-Ended (Differential) Input Mode

Two separate signals V_{i1} and V_{i2} are applied between the two inputs; the op-amp amplifies their difference **$V_d = V_{i1} - V_{i2}$**.

Double-Ended (Differential) Output Mode

Output is measured as a floating voltage between two output terminals V_{o1} and V_{o2} (neither is grounded). Because V_{o1} and V_{o2} are equal in amplitude and 180° out of phase, the differential output signal **$V_{o1} - V_{o2}$ is twice the amplitude (2×)** of either single-ended output!

Common-Mode Operation & BJT Differential Amplifier Pair

When the exact same signal (e.g., 60 Hz / 50 Hz power-line hum or thermal drift) appears simultaneously on both inputs ($V_{i1} = V_{i2}$), an ideal differential amplifier produces zero output. Internally, this is achieved by the symmetric **BJT Differential Pair (Q_1, Q_2 sharing a common emitter tail resistor R_E connected to $-V_{EE}$)**.

5. Internal 4-Stage

[Input Signals V_1, V_2] → [1. Input Stage] → [2. Intermediate Stage] → [3. Level Shifting Circuit] → [4. Output Stage] → [Output V_o]

Stage 1 — Input Stage (Dual-Input, Balanced-Output Differential Amplifier): Provides most of the voltage gain, establishes the very high input resistance R_i, and rejects common-mode noise.

Stage 2 — Intermediate Stage (Dual-Input, Unbalanced Single-Ended Output Differential Amplifier): Driven directly (DC-coupled) by Stage 1 to provide

	additional high voltage gain. **Stage 3 — Level Shifting Circuit (Emitter Follower with DC Voltage Divider):** Because Stages 1 and 2 are direct-coupled (no coupling capacitors), the DC quiescent voltage builds up well above 0 V. The Level Shifter shifts the DC quiescent level back down to **exact 0 V with respect to ground** when input is zero, while providing high input impedance and low output impedance. **Stage 4 — Output Stage (Complementary Symmetry Push-Pull Amplifier):** Increases the output voltage swing to near $\pm V_{CC}$, increases current-sourcing capability, and provides very **low output resistance R_o (≈ 75 Ω in IC 741, 0 Ω ideal)**.
Ideal Op-Amp vs. Practical IC 741 (5 Specs)	(1) Open-loop Voltage Gain A_{OL}: **∞** vs. **10^5 (100 dB)**; (2) Input Impedance R_{in}: **∞** vs. **2 MΩ**; (3) Output Impedance R_{out}: **0 Ω** vs. **75 Ω**; (4) Bandwidth: **∞** vs. **1 MHz**; (5) CMRR (Common-Mode Rejection Ratio = $\lvert A_d / A_{cm} \rvert$): **∞** vs. **90 dB**; Slew Rate (max dV_{out}/dt) = **0.5 V/μs**.
Inverting Amplifier (Figure 2.4-1)	V_+ is grounded (0 V), so by Virtual Ground **$V_- = 0$ V**. KCL at (−): $V_{in}/R_1 + V_{out}/R_f = 0 \Rightarrow$ **$A_v = V_{out} / V_{in} = -(R_f / R_1)$** (Amplifies with 180° phase inversion).
Non-Inverting Amplifier (Figure 2.4-2)	V_{in} is applied to V_+, so **$V_- = V_{in}$**. KCL at (−): $(V_{in} - 0)/R_1 + (V_{in} - V_{out})/R_f = 0 \Rightarrow$ **$A_v = V_{out} / V_{in} = 1 + (R_f / R_1)$** (0° phase shift; setting $R_f = 0$, $R_1 = \infty$ gives **Voltage Follower $V_{out} = V_{in}$**).
Inverting Summing Amplifier (Figure 2.4-3)	KCL at virtual-ground (−) node gives **$V_{out} = -R_f \cdot (V_1/R_1 + V_2/R_2 + V_3/R_3)$**.

M2T7 Module 2, topic 7

Electromagnetic Spectrum, Communication Block Diagram & Modulation

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	Every electromagnetic wave travels at the speed of light **$c = 3 \times 10^8$ m/s = $f \cdot \lambda$**. Look at the **Shannon-Weaver Communication System Block Diagram** and **AM vs. FM Waveforms** in **Figure 2.5 (Panels 1 & 2)**:
ITU Radio Frequency Bands & Propagation Modes	**VLF/LF/MF (3 kHz–3 MHz, Ground Wave** hugging Earth's curvature for AM radio); **HF (3–30 MHz, Sky Wave** bouncing off the Ionosphere for international shortwave); **VHF (30–300 MHz, Line-of-Sight** for FM radio & TV); **UHF (300 MHz–3 GHz, Line-of-Sight** for 4G/5G Mobile, Wi-Fi, GPS); **SHF/EHF (3–300 GHz** for Satellite, RADAR, and 5G mmWave).
Why Baseband Signals MUST Be Modulated (Classic 5-Mark Proof)	A human voice signal (f = 3 kHz) has wavelength **$\lambda = c / f = (3 \times 10^8) / (3 \times 10^3) =$ 100,000 m (100 km)**. Since an efficient dipole antenna must have length **$L = \lambda / 4$**, an unmodulated 3 kHz voice wave would require a **25 km tall vertical antenna tower** (impossible!). By modulating a 100 MHz carrier (λ = 3 m), antenna size drops to **$L = 3 / 4 = 0.75$ m**! Modulation also allows **Multiplexing (FDM)** so thousands of stations don't mix in the air.

AM vs. FM vs. PM	In **AM (Amplitude Modulation)**, carrier amplitude varies linearly with message m(t) while frequency stays constant (simple, narrow bandwidth, but vulnerable to lightning/spark noise). In **FM (Frequency Modulation)** and **PM (Phase Modulation)**, carrier amplitude stays 100% constant while frequency/phase varies—making FM immune to amplitude noise!

M2T8 Module 2, topic 8

Cellular Communication Systems (Hexagons, Reuse, Handoff & 1G to 5G)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	How do millions of people in a city talk on mobile phones using a limited slice of radio spectrum? By dividing the city into small **Hexagonal Cells** (each served by a low-power **Base Transceiver Station, BTS** connected to a **Mobile Switching Center, MSC**, Figure 2.5-3)!
Why Hexagonal Cells Instead of Circles or Squares?	Circular coverage footprints either leave dead-zone gaps or overlap when tiled; among regular polygons that tessellate a flat plane without gaps (equilateral triangle, square, regular hexagon), a **Hexagon is closest to a circle** and covers the **largest area** for a given antenna transmission radius R!
Frequency Reuse & Cluster Size (N)	A group of **N** adjacent cells that use the entire allocated frequency spectrum once (with different channel sets A, B, C, D, E, F, G) is called a **Cluster**: $\mathbf{N = i^2 + i \cdot j + j^2}$ (so valid cluster sizes are N = 1, 3, 4, 7, 9, 12, 13...). If each cell has *k* channels and the cluster of *N* cells is repeated *M* times across a city, total system capacity is $\mathbf{C = M \cdot k \cdot N}$ (Frequency Reuse Factor = **1 / N**).
Handoff (Handover) vs. Roaming	**Handoff** is transferring an active call from one cell tower (BTS) to the next as a car moves across a cell boundary (*Hard Handoff* = Break-Before-Make in 2G GSM; *Soft Handoff* = Make-Before-Break in 3G CDMA). **Roaming** is moving outside your home operator's network coverage into another operator's network (tracked via Home Location Register HLR & Visitor Location Register VLR).
Evolution from 1G to 5G	**1G (1980s):** Analog voice only (FDMA, AMPS, 2.4 kbps). **2G (1990s):** Digital voice + SMS text (GSM/TDMA & CDMA, 14.4–64 kbps). **3G (2000s):** Mobile web browsing & video calling (WCDMA / HSPA, 384 kbps–2 Mbps). **4G (2010s):** All-IP packet network, HD streaming & VoLTE (OFDMA + MIMO, 100 Mbps–1 Gbps). **5G (2020s):** 3 pillars—**eMBB** (up to 20 Gbps), **uRLLC** (< 1 ms latency for autonomous cars/remote surgery), and **mMTC** (1 million IoT devices/km^2 using mmWave & Massive MIMO beamforming).

M2T9 Module 2, topic 9

Fundamentals of Satellite Communication (Orbits & Uplink vs. Downlink)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	A communication satellite is an active **Microwave Repeater (Transponder)** in space that receives weak signals from an Earth Station on the **Uplink frequency**, shifts the frequency via a local oscillator mixer, amplifies it with a High-Power

Amplifier (TWTA), and beams it back to Earth on the **Downlink frequency** (**Figure 2.6-1**).

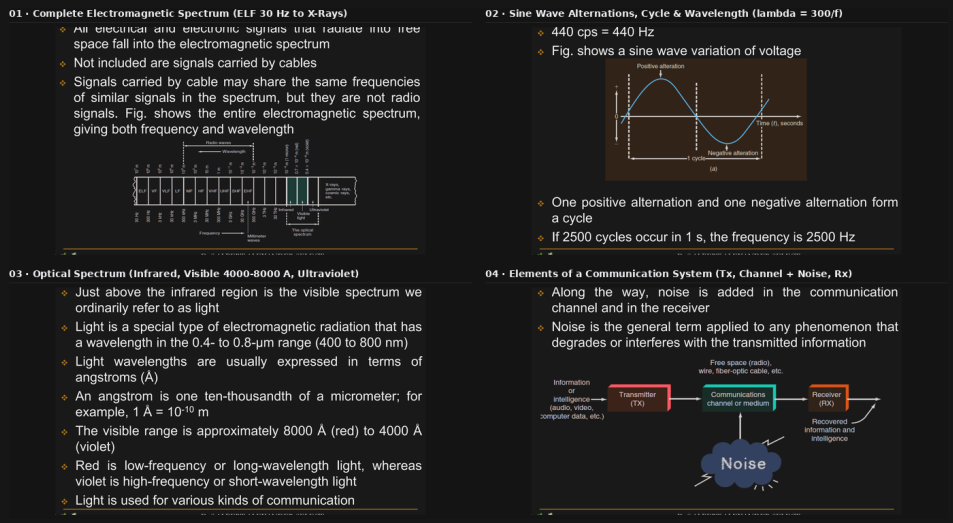


Figure 2.5B (Extracted from VIT Lecture Slides): (a) Complete Electromagnetic Spectrum (30 Hz ELF to X-Rays), (b) Sine Wave Alternations & Wavelength ($\lambda = 300/f_{MHz}$), (c) Optical Spectrum (IR, Visible 4000–8000 Å, UV), (d) Elements of a Communication System with Channel Noise.

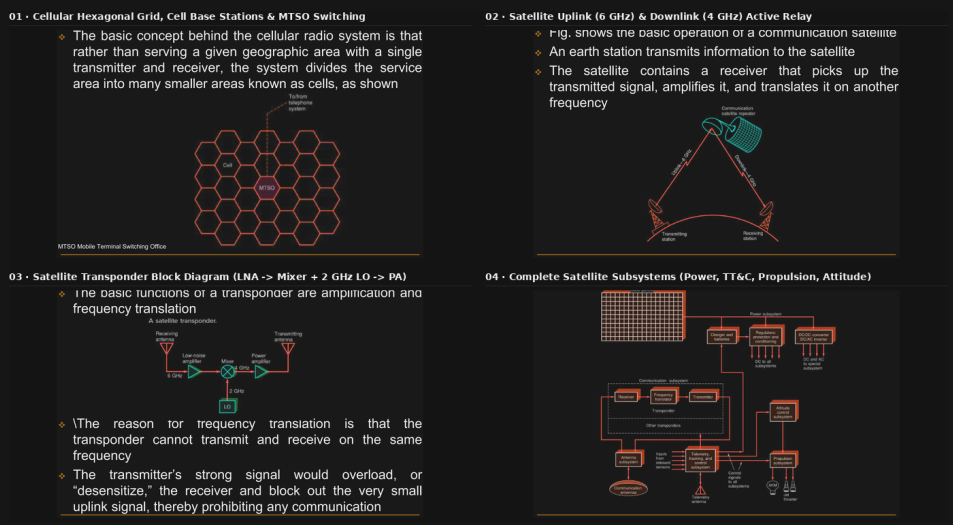


Figure 2.6B (Extracted from VIT Lecture Slides): (a) Cellular Hexagonal Grid & MTSO Link, (b) Satellite Uplink (6 GHz) & Downlink (4 GHz), (c) Satellite Transponder Internal Block Diagram (LNA → Mixer + 2 GHz LO → PA), (d) Complete Satellite Subsystems Block Diagram.

Complete Slide Details

Frequency & Wavelength

Frequency f is the number of cycles per second (**1 cps = 1 Hz**, named after Heinrich Hertz; 1 positive alternation + 1 negative alternation = 1 complete cycle; household AC = 50/60 Hz; musical note A = 440 Hz). Metric prefixes: **kHz (10^3), MHz (10^6), GHz (10^9), THz (10^{12})**. Since radio waves travel at the speed of light **c = 3×10^8 m/s (186,000 miles/s)**, wavelength is **λ (m) = 300,000,000 / f (Hz) = 300 / f (MHz)**.

Optical Spectrum & Ångström Unit (Slide 18–20 of EM Deck)

Between 300 GHz and the optical region lies a largely uninhabited millimeter/sub-millimeter gap. Above it lies the **Optical Spectrum** (measured by wavelength in μm, nm, or **Ångströms where 1 Å = 10^{-10} m**):

	Infrared (IR): 0.1 mm to 700 nm (100 µm to 0.7 µm) — emitted by warm/heated objects; used in TV remote controls, thermal imaging, astronomy, missile guidance, and **fiber-optic communication**. **Visible Light: 0.4 µm to 0.8 µm (400 nm to 800 nm)**, i.e., from **4000 Å (Violet) to 8000 Å (Red)** — used in free-space laser communication and optical fibers. **Ultraviolet (UV): 4 nm to 400 nm** — generated by mercury vapor lamps and the sun; used in medical sterilization (not used for communication); above UV lie X-rays, Gamma rays, and Cosmic rays.
Bandwidth (BW = f2 − f1) & Special Communication Channels	Bandwidth is the width of the frequency range occupied by an information signal or channel (e.g., human voice VF from f_1 = 300 Hz to f_2 = 3000 Hz has **BW = 3000 − 300 = 2700 Hz = 2.7 kHz**). In addition to twisted-pair wire, coaxial cable, fiber-optic "light pipes", and free space, the slides highlight three special communication media: 1. **Water (Sonar — SOund Navigation And Ranging):** Used by submarines and surface ships because water conducts acoustic sound waves well; *Passive Sonar* listens to underwater propeller/engine noise using **hydrophones** (underwater microphones), while *Active Sonar* transmits acoustic pulses and detects echoes from hulls, schools of fish, or the ocean floor. 2. **The Earth:** Conducts low-frequency seismic waves and ground-current signals. 3. **AC Power Lines (Power Line Carrier / PLC):** Household and high-voltage utility 50/60 Hz power lines carry superimposed high-frequency carrier signals for intercoms, smart meters, and substation telemetry!
Complete Slide Details	
AMPS & Digital Generations	Pioneered by **Bell Laboratories / AT&T** in the 1970s and deployed commercially in the early 1980s. First-generation (1G) **AMPS (Advanced Mobile Phone System)** used analog FM; today replaced by 2G/3G/4G/5G digital systems that fit more channels into a smaller spectrum and carry voice, text, images, and internet data.
Hexagonal Cells & MTSO Handoff	Each cell covers a few square miles (smaller cells with shorter towers and lower transmit power in dense cities; larger cells with tall towers in rural areas) and links via fiber/microwave to the **MTSO (Mobile Telephone Switching Office)**, which connects to the public switched telephone network (PSTN) and manages automatic **Handoff** when a moving vehicle's signal drops below threshold.
Exact FCC Cellular	**Original 800–900 MHz Cellular Band (formerly UHF TV Channels 68–83):** Two 25-MHz bands providing 832 channels (30 kHz each): **Uplink (Mobile-to-Base Reverse Channel): 824 – 849 MHz**; **Downlink (Base-to-Mobile Forward Channel): 869 – 894 MHz** (duplex spacing = 45 MHz). **Extended 900 MHz Band (GSM-900): Uplink: 890 – 915 MHz (25 MHz)**; **Downlink: 935 – 960 MHz (25 MHz)**. **PCS (Personal Communications Systems — 1.9 GHz Digital Band):** Two 60-MHz blocks: **Uplink: 1850 – 1910 MHz**; **Downlink: 1930 – 1990 MHz** (plus 700–800 MHz UHF digital TV transition spectrum).
Complete Slide Details	
Why Satellites Translate Frequency (Transponder	A communication satellite is an orbiting RF repeater/relay station that overcomes Earth's curvature line-of-sight limitation. In a C-band transponder, the **6 GHz**

Block Diagram)	**Uplink** signal from the Earth station is picked up by the Receiving Antenna, amplified by a **Low-Noise Amplifier (LNA)**, fed into a **Mixer driven by a 2 GHz Local Oscillator (LO)** to down-convert the carrier to **6 – 2 = 4 GHz**, amplified by a high-power **Power Amplifier (TWTA/SSPA)**, and beamed back to Earth as the **4 GHz Downlink**. *Why translate from 6 GHz to 4 GHz?* Because if the satellite transmitted and received on the exact same frequency, the blazing multi-watt output of its own Power Amplifier would leak into and completely overload/desensitize its ultra-sensitive microwatt LNA receiver!
Dual Polarization Frequency Reuse	A standard satellite carries 12 transponders (36 MHz bandwidth each), and by using **orthogonal vertical and horizontal linear polarization (or LHCP/RHCP circular polarization)**, two transponders share the exact same frequency without interference—doubling capacity to **24 channels (12 vertically polarized + 12 horizontally polarized)**!
Complete Satellite Letter Band Table (Slide 9 of Satellite Deck)	**P Band:** 225 – 390 MHz | **J Band:** 350 – 530 MHz | **L Band:** 1530 – 2700 MHz (GPS navigation, Inmarsat marine & aeronautical mobile) | **S Band:** 2500 – 2700 MHz. **C Band (3400 – 6425 MHz):** Most widely used band; **Uplink: 5.925 – 6.425 GHz (~6 GHz)**, **Downlink: 3.7 – 4.2 GHz (~4 GHz)**. Since earthly point-to-point microwave links also share 4/6 GHz, Earth stations must be sited carefully; Immune to rain fade! **X Band:** 7250 – 8400 MHz (Military & defense satellites). **Ku Band (10.95 – 14.5 GHz): Uplink: 14.0 – 14.5 GHz (~14 GHz)**, **Downlink: 11.7 – 12.2 GHz (~12 GHz)**. Avoids terrestrial microwave interference and gives higher antenna gain for a given reflector size, allowing compact DTH rooftop dishes! **Ka Band:** 17.7 – 31 GHz (30/20 GHz high-throughput broadband) | **Q Band:** 36 – 46 GHz | **V Band:** 46 – 56 GHz | **W Band:** 56 – 100 GHz.
Satellite Subsystems Block Diagram (Slide 16 of Satellite Deck)	Every communication satellite comprises the **Communication Payload (Transponders + Antennas)** supported by 5 essential **Housekeeping Subsystems**: (1) **Power Subsystem** (Solar cell arrays charging backup NiCd/Li-ion batteries during solar eclipse via charge regulators and DC-to-DC / DC-to-AC converters), (2) **Telemetry, Tracking & Command (TT&C)** (monitors spacecraft temperatures/voltages, tracks orbital range/velocity, and executes ground commands), (3) **Propulsion Subsystem** (**AKM — Apogee Kick Motor** injects satellite from elliptical transfer orbit into circular 35,786 km GEO orbit, while hydrazine **Jet Thrusters** perform N-S and E-W station-keeping), (4) **Attitude Control Subsystem** (spin stabilization or 3-axis momentum wheels + horizon sensors keep directional antennas pointed at Earth), and (5) **Antenna Subsystem** (horn, parabolic spot-beam, and omnidirectional TT&C antennas).

Why is Uplink	**Why is Uplink Frequency (6 GHz C-band / 14 GHz Ku-band) Always HIGHER than Downlink Frequency (4 GHz / 11 GHz)?** Free-space and atmospheric rain attenuation increase with the square of frequency (**Loss ∝ f^2**). An Earth Station on the ground has access to unlimited AC grid power and massive parabolic dishes to blast a 6 GHz signal upward against high path loss. The satellite in space runs on limited solar panels and batteries, so it transmits downward at a **lower frequency (4 GHz)** that suffers less atmospheric loss! Also, using two different frequencies prevents the satellite's powerful transmitter from jamming its own ultra-sensitive receiver!
The 3 Satellite Orbits (LEO vs. MEO vs. GEO)	(1) **LEO (Low Earth Orbit, 500–2,000 km, period ~90–120 min):** Lowest signal delay (~5 ms) and lowest launch cost, used for Earth imaging and **Starlink / Iridium** constellations. (2) **MEO (Medium Earth Orbit, 8,000–20,200 km, period ~12 hours):** Used for **GPS / NavIC / Galileo navigation** (requires simultaneous distance pseudoranges from at least **4 satellites** to solve for 3D position X, Y, Z plus receiver clock bias Δt!). (3) **GEO (Geostationary Orbit, 35,786 km above the Equator, period = 24 hours):** Orbits at the exact rotational speed of Earth so it appears fixed at one spot in the sky—dish antennas (DTH TV, INSAT weather) never need to track! Just **3 GEO satellites** spaced 120° apart cover almost the entire globe.

M2T10 Module 2, topic 10

Fundamentals of RADAR Systems (Block Diagram, Range Equation & Doppler)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	**RADAR (Radio Detection and Ranging)** determines a target's distance (range), angle (azimuth/elevation), and radial velocity by transmitting high-power microwave pulses and timing the reflected echo (**Figure 2.6-2**):
Role of the Duplexer	A fast microwave T/R (Transmit/Receive) switch that connects the **Transmitter** to the single antenna during the microsecond pulse (blocking and protecting the delicate receiver from burning out!), then instantly flips the antenna to the **Receiver** to listen for the weak nanowatt echo!
Round-Trip Target Range & Maximum Unambiguous Range	Because the pulse travels to the target and back (total distance = 2R) at speed of light $c = 3 \times 10^8$ m/s, target distance is $\mathbf{R = (c \cdot \Delta t) / 2}$. To prevent echoes from arriving after the next pulse is sent (Second-Time-Around Echoes), Maximum Unambiguous Range is $\mathbf{R_{unamb} = c / (2 \cdot PRF) = (c \cdot T_R) / 2}$, where PRF is Pulse Repetition Frequency.
Free-Space Radar Range Equation	$\mathbf{R_{max} = [(P_t \cdot G^2 \cdot \lambda^2 \cdot \sigma) / ((4\pi)^3 \cdot P_{r,min})]^{1/4}}$, where P_t = peak transmit power, G = antenna gain, λ = wavelength, σ = Radar Cross Section (RCS in m^2), and $P_{r,min}$ = minimum detectable signal. Notice the 1/4th root: to **double** radar range R_{max}, you must increase transmit power P_t by $\mathbf{2^4 = 16}$ **times**!
Continuous-Wave (CW) Doppler Radar	Uses the **Doppler Effect**—when a target moves toward or away from the radar at radial speed v_r, the reflected frequency shifts by $\mathbf{f_d = (2 \cdot v_r) / \lambda = (2 \cdot v_r \cdot f_0) / c}$

(used in police speed guns, MTI air-traffic control, and missile seekers).

M2T11 Module 2, topic 11

Module 2 FAT PYQ Self-Assessment & Multi-Concept Solved Problems

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context

Test your topic-by-topic mastery on these official VIT CAT-2 / FAT questions (VIT CAT-2 E1 Paper & Official Key).

PYQ 1 [10 Marks — Multi-Concept Power Supply]: Bridge Rectifier + Zener Regulator Combined Problem

Overview & context

Question: A 230 V, 50 Hz AC mains feeds a 10:1 step-down transformer connected to a Silicon Bridge Full-Wave Rectifier (V_D = 0.7 V per diode). (a) Calculate the peak load voltage $V_{m(load)}$ and average DC output V_{dc} across a 100 Ω load. (b) If a smoothed 24 V DC bus from a filter feeds a 10 V Zener shunt regulator via R_S = 280 Ω connected to a load R_L = 500 Ω, find I_S, I_L, I_Z, and Zener power dissipation P_Z.

Part (a) Bridge Rectifier (5 Marks)

$V_{2(rms)}$ = 230 / 10 = 23 V ⇒ V_m = 23√2 = **32.53 V**. Subtracting 2 series silicon diodes (2 × 0.7 = 1.4 V): **$V_{m(load)}$ = 32.53 − 1.4 = 31.13 V**. Average DC output: **V_{dc} = $2V_{m(load)}$ / π = (2 × 31.13) / 3.1416 = 19.82 V**; I_{dc} = 19.82 / 100 = **198.2 mA**; I_{rms} = (31.13 / 100) / √2 = **220.1 mA**.

Part (b) Zener Regulator (5 Marks)

Series current: **I_S = (24 − 10) / 280 = 0.05 A (50 mA)**. Load current: **I_L = 10 / 500 = 0.02 A (20 mA)**. Zener current: **I_Z = I_S − I_L = 50 − 20 = 30 mA**. Zener power: **P_Z = V_Z · I_Z = 10 V × 30 mA = 300 mW (0.3 W)**.

PYQ 2 [10 Marks — VIT CAT-2 E1]: BJT Gains & Multi-Stage Op-Amp Summing + Non-Inverting Cascade

Overview & context

Question: (a) An NPN transistor in CE configuration has α = 0.985 and collector current I_C = 3.94 mA. Find β, base current I_B, and emitter current I_E. (b) Three inputs V_1 = +2 V, V_2 = −3 V, V_3 = +4 V feed an inverting summing Op-Amp via R_1 = 20 kΩ, R_2 = 30 kΩ, R_3 = 40 kΩ with R_f = 60 kΩ. If its output feeds a Non-Inverting Op-Amp with R_1 = 10 kΩ and R_f = 15 kΩ, find the final output voltage!

Part (a) BJT Numerical (4 Marks)

β = α / (1 − α) = 0.985 / 0.015 = 65.67. Base current: **I_B = I_C / β = 3.94 mA / 65.67 = 0.06 mA (60 µA)**. Emitter current: **I_E = I_B + I_C = 0.06 + 3.94 = 4.00 mA**.

Part (b) Cascaded Op-Amp Numerical (6 Marks)

Stage 1 (Inverting Summer): **V_{out1} = −R_f (V_1/R_1 + V_2/R_2 + V_3/R_3) = −60 (2/20 − 3/30 + 4/40) = −60 (0.1 − 0.1 + 0.1) = −6.0 V**. Stage 2 (Non-Inverting Amplifier): Gain **A_{v2} = 1 + (R_f / R_1) = 1 + (15 / 10) = 2.5**. Final output: **V_{final} = 2.5 × (−6.0 V) = −15.0 V**!

PYQ 3 [10 Marks — VIT FAT Wireless]: Cellular Capacity, Antenna Height & Pulsed/Doppler RADAR

Overview & context

Question: (a) Calculate the minimum quarter-wave antenna height required to radiate a 5 kHz audio signal directly versus a 150 MHz VHF carrier. (b) A cellular network with 42 MHz spectrum (50 kHz per duplex channel) uses a 7-cell cluster (i=2, j=1) repeated 12 times. Find k and total capacity C. (c) A 10 GHz police

	Doppler radar (λ = 0.03 m) measures an echo after Δt = 4 μs with a Doppler shift f_d = 2,000 Hz. Find target range R and speed v_r.
Part (a) Antenna Height (3 Marks)	At 5 kHz: $\lambda_1 = (3\times10^8)/(5\times10^3)$ = 60,000 m ⇒ **$L_1 = \lambda_1/4$ = 15,000 m (15 km — impractical!)**. At 150 MHz: $\lambda_2 = (3\times10^8)/(1.5\times10^8)$ = 2 m ⇒ **$L_2 = \lambda_2/4$ = 0.5 m (practical!)**.
Part (b) Cellular Capacity (3 Marks)	Total channels = 42,000 kHz / 50 kHz = 840. With N = 7: **k = 840 / 7 = 120 channels/cell**. Total capacity across M = 12 clusters: **C = 12 × 120 × 7 = 10,080 simultaneous calls**.
Part (c) Pulsed & Doppler Radar (4 Marks)	Range: **$R = (c \cdot \Delta t) / 2 = (3\times10^8 \times 4\times10^{-6}) / 2 = 600$ m**. From $f_d = 2v_r / \lambda$: **$v_r = (f_d \cdot \lambda) / 2 = (2{,}000 \times 0.03) / 2 = 30$ m/s (108 km/h)**!

PYQ 4 [10 Marks — VIT FAT Devices & Space Systems]: BJT vs. MOSFET & Satellite Orbit / Link Design

Overview & context	**Question:** (a) Compare a Bipolar Junction Transistor (BJT) and a Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) across six engineering parameters, and explain why an Enhancement MOSFET has nearly infinite input impedance. (b) Compare LEO, MEO, and GEO satellite orbits, explain why Uplink frequency (6 GHz) is always kept higher than Downlink frequency (4 GHz), and state why a GPS receiver requires signals from at least 4 MEO satellites simultaneously.
Part (a) BJT vs. MOSFET (5 Marks)	(1) **Charge Carriers:** BJT is *bipolar* (both electrons and holes conduct); MOSFET is *unipolar* (only majority carriers conduct). (2) **Control Parameter:** BJT is a *current-controlled* device ($I_C = \beta I_B$); MOSFET is a *voltage-controlled* device (I_D controlled by V_{GS}). (3) **Input Impedance (Z_{in}):** BJT has low Z_{in} (~1–2 kΩ) because its Base-Emitter junction is forward-biased; MOSFET has **nearly infinite Z_{in} (~10^{12} Ω)** because a thin **SiO_2 glass insulating layer** blocks any DC gate current ($I_G \approx 0$ A). (4) **Thermal Stability:** BJT has a positive temperature coefficient of collector current and suffers from *thermal runaway*; MOSFET has a negative temperature coefficient at high currents and is immune to thermal runaway. (5) **Switching Speed & IC Density:** MOSFET switches much faster (no minority carrier storage delay) and occupies a tiny area on VLSI microchips.
Part (b) Satellite Orbits, Uplink/Downlink & GPS (5 Marks)	(1) **Orbits:** LEO (500–2,000 km, 90–120 min period, low latency for Starlink/imaging); MEO (8,000–20,200 km, 12 hr period for GPS); GEO (35,786 km equatorial, 24 hr period, stationary in sky for DTH TV & weather). (2) **Why Uplink (6 GHz) > Downlink (4 GHz):** Free-space path loss and rain attenuation increase with frequency squared ($L \propto f^2$). Ground Earth stations have unlimited grid power to overcome 6 GHz attenuation, whereas satellites rely on limited solar arrays and use 4 GHz for lower downlink loss. (3) **Why 4 GPS Satellites:** Four simultaneous pseudorange equations are required to solve for four unknowns: 3D spatial coordinates (**Latitude X, Longitude Y, Altitude Z**) plus the **Receiver Clock Bias (Δt)**!